PRAHARI

THE COUNTDOWN BAND

The hazard outside. The strain inside.
One warning on your wrist.

PROBLEM STATEMENT ID 26181

PROBLEM STATEMENT TITLE

A secure, AI-powered Personal Health Companion that delivers real-time, privacy-preserving health monitoring and early warning capabilities, helping individuals recognize health risks before they become emergencies. The solution should improve resilience during heat waves, floods, pollution events, and other disasters common in India while enabling continuous health support through on-device intelligence.

THEME
MedTech / BioTech / HealthTech

PS CATEGORY
Hardware

TEAM ID
To be provided



Built for people working beyond the last signal.

Photo: yajna S R / Pexels · Mumbai, India

Two threats. One personal warning system.

Proposed Solution / PRAHARI — The Countdown Band

01 THE HAZARD OUTSIDE

Warnings must travel past a network outage.

Official CAP alerts → phone + gateway band
→ LoRa peer relay → vibration at the wrist

02 THE STRAIN INSIDE

A regional forecast cannot see your body.

Heart rate + skin temperature + ambient heat
→ on-device trend → early rest / cooling prompt

THE BAND

Senses • decides • warns

THE COMPANION APP

Fetches alerts • keeps offline history

ONE CONNECTED ENTRY POINT
SHARES WARNINGS LOCALLY

INNOVATION & UNIQUENESS

*Proposed features; personal forecasts and recovery tracking require validation.

Body + hazard in one device

Time to heat-risk threshold*

Recovery across shifts*

Sense locally. Decide on-device. Relay the warning.

Proposed rev B architecture · hardware + embedded intelligence + offline companion

1 SENSE

MAX30102 · heart rate / PPG
MAX30208 · contact skin temp
AHT20 · air temp + humidity
Onboard IMU · motion context

2 DECIDE

Signal-quality + motion gate
Kalman estimate + personal baseline
Trend-to-threshold forecast*
Recovery / thermal-debt ledger*

3 WARN

Distinct haptics + OLED status
SX1262 LoRa · local peer relay*
BLE · Android companion*
Local records + protected keys*



XIAO nRF52840 Sense

C/C++ · Arduino | TinyML planned
Flutter / Dart · local storage planned

Privacy by design

Keep raw physiology on-device;
share only necessary alert data.*

CAP → connected phone → BLE gateway band → LoRa peers → haptic warning

Board image: Seeed Studio · *Planned / requires validation; air temperature + humidity alone are not full WBGT.

FEASIBILITY AND VIABILITY

A working sensing core. A focused path to a field pilot.

FEASIBILITY TODAY

Verified on the bench

Live heart-rate sensing + OLED
I²C discovery and sensor diagnostics

Engineering in progress

Rev B PCB + BOM available
Radio bring-up / PCB checks open
App + forecasting + relay to integrate

CHALLENGE	STRATEGY TO OVERCOME IT
Motion & sensor uncertainty	Quality gate; reference-device validation; show “estimate unavailable” on poor signal.
Radio loss & stale alerts	Time stamps, expiry, deduplication, hop limits; measure delivery and battery life in the field.
Fit, power & deployment cost	Modular parts; duty cycling; enclosure trials; quote the complete rev B assembly before pilot.

Pilot readiness gates

01 **Close hardware issues**

02 **Validate sensing + relay**

03 **Supervised worker pilot**

Economic viability: off-the-shelf modules + a two-layer PCB; final cost awaits rev B BOM, battery, enclosure and assembly quotes.

Earlier awareness. A clearer next action.



Start with outdoor work crews.

Construction · agriculture · municipal work

Photo: yajna S R / Pexels

57% of Indian districts face high to very high heat risk.

CEEW, 2025 · district-level heat-risk assessment [1]

SOCIAL / Reach the person

Simple haptics; no personal smartphone needed for local warning reception.

ECONOMIC / Support timely action

Earlier rest and cooling prompts aim to reduce disruption; pilot outcomes will test the benefit.

MEASURE IN THE PILOT Alert delivery time • Missed / false alerts • Time to action • Shift-long wearability

Grounded in research. Built to be tested.

Reference & research work · clickable source titles

01 THE NEED

CEEW · How Extreme Heat is Impacting India (2025)

District-level heat risk; basis for the 57% figure.

02 OFFICIAL WARNING SOURCE

NDMA · SACHET National Disaster Alert Portal

CAP-based multi-hazard alerts and RSS dissemination.

03 ALGORITHM FOUNDATION

Buller et al. · Wearable physiological estimation

Kalman-filter research; PRAHARI still needs its own validation.

04 HARDWARE PLATFORM

Seeed Studio · XIAO nRF52840 Series

MCU, BLE and Sense IMU documentation; module image source.

PHOTOGRAPHY yajna S R / Pexels · Mumbai construction workers · Pexels License

PROJECT EVIDENCE

PRAHARI repository

Sensor + OLED firmware
KiCad design + BOM
Bring-up and verification notes

Obsidian project brain
PRAHARI reference deck + report

Repository access may be restricted.